

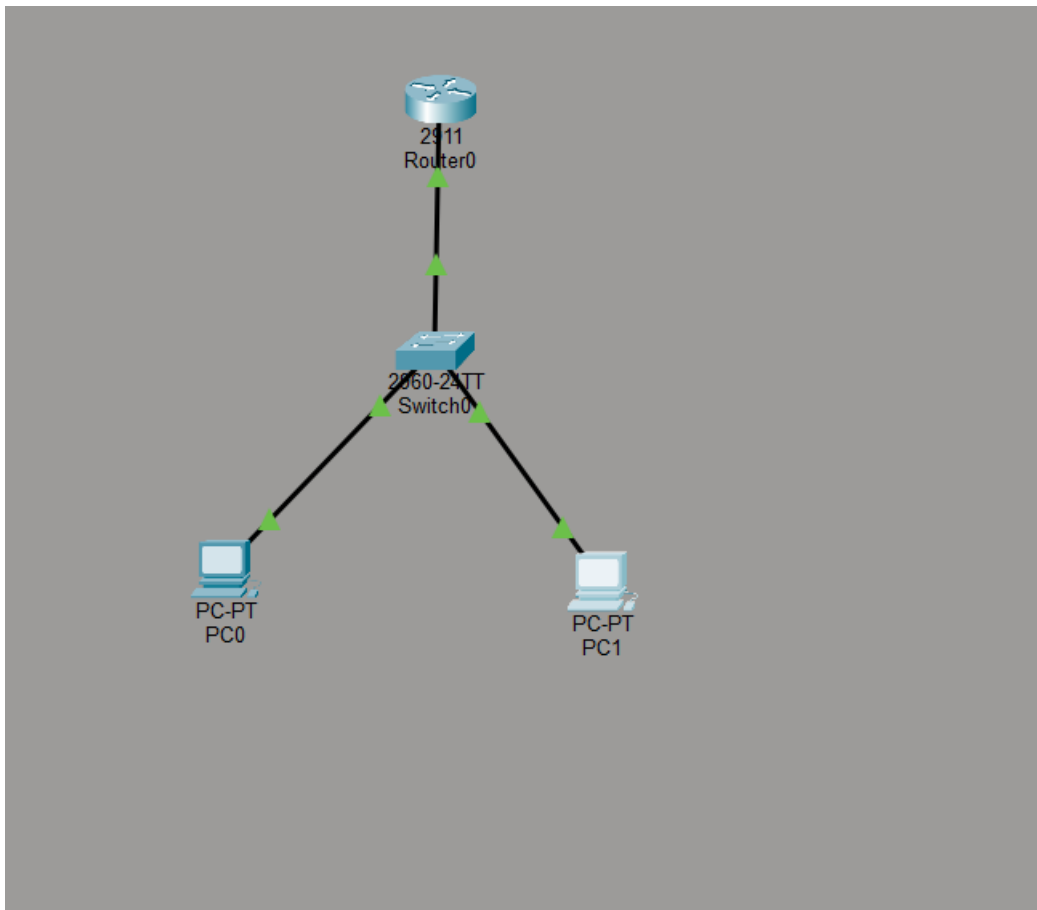
Q-Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

To configure a router as a DHCP server with only one available IP address and observe that only one client PC receives an IP address while the second PC fails to obtain one.

Requirements

- Cisco Packet Tracer
- 1 Router (2911/1941)
- 1 Switch (2960)
- 2 PCs
- Copper Straight-Through Cables

Network Topology



Router Configuration

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#ip dhcp excluded-address 192.168.1.3 192.168.1.254
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#show running-config
^
% Invalid input detected at '^' marker.

Router(config)#show running-config
^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show running-config
Building configuration...

Current configuration : 857 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
ip dhcp excluded-address 192.168.1.3 192.168.1.254
!
ip dhcp pool LAN
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
```

Procedure

1. Added one router, one switch, and two PCs in Cisco Packet Tracer.
2. Connected all devices using Copper Straight-Through cables.
3. Configured the router interface with the IP address **192.168.1.1/24**.
4. Configured the router as a DHCP server.
5. Excluded all IP addresses except **192.168.1.2**, leaving only one IP available for DHCP.
6. Configured both PCs to obtain IP addresses automatically using DHCP.
7. Verified the assigned IP addresses using the **ipconfig** command.

PC0 DHCP :

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

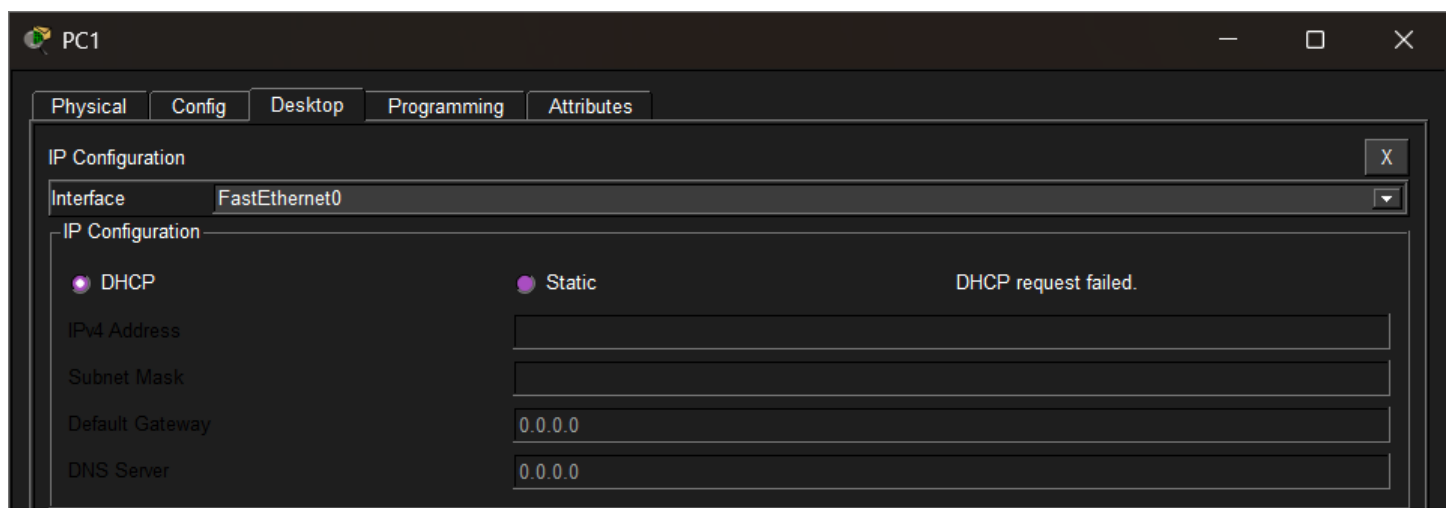
    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:A3FF:FEAA:73D8
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```

PC1 DHCP :



Only one IP address was available in the DHCP pool. The first PC successfully obtained that IP address through DHCP, while the second PC could not receive an IP address because the DHCP pool had no remaining addresses to assign.